Inclusion–exclusion solutions

1 • Three divisibility conditions

The universe is {1,…,200}. Single bad sets have sizes 100,66,40. Pair intersections use lcm 6,10,15 and have sizes 33,20,13. The triple intersection consists of multiples of 30, size 6. Avoiding all gives

$$200 - (100 + 66 + 40) + (33 + 20 + 13) - 6 = 54.$$

These are actual intersections in one finite universe. The probability 33/200 of divisibility by both 2,3 differs from (100/200)(66/200). Accidental equality for one pair would not establish mutual independence either.

2 • Exactly two owners recover their passes

Choose the two fixed owners in $\binom{6}{2}$ ways and derange the remaining four, so no other owner receives the correct pass. Distinct fixed sets give disjoint classes; every target assignment has a unique fixed set. Since $D_4=9$, the answer is $\binom{6}{2} D_4 = 15 \cdot 9 = 135$. Using arbitrary 4! assignments for the rest would also count three or more fixed points.

3 • Two checks for derangements

For $n \geq 2$ fix $\pi(1)=j \neq 1$, with $n-1$ choices. If $\pi(j)=1$, remove the two-cycle (1 j), leaving a derangement on the other $n-2$ labels; restoring the cycle reverses the operation. Otherwise let k be 1's predecessor, with $k \neq j$. Contract $k \to 1 \to j$ to $k \to j$. No new fixed point occurs because that would require $k=j$; all other edges are unchanged. Conversely insert 1 before j at its unique predecessor in any derangement of the remaining $n-1$ labels. The cases are disjoint, exhaustive, and reversible, proving

$$D_n = (n-1)(D_{n-1} + D_{n-2}).$$

The empty set has one empty permutation, so $D_0=1$; $D_1=0$. Then $D_2=1, D_3=2, D_4=9, D_5=4(9+2)=44, D_6=5(44+9)=265$. Inclusion–exclusion gives $D_5=120-120+60-20+5-1=44$. Setting the empty base to zero would break the recurrence at n=2.

4 • A full forbidden board

Choose k rows, k columns, and their bijection: $r_k = \binom{3}{k}^2 k!$. Coefficients are 1,9,18,6, so $R_B = 1+9x+18x^2+6x^3$. Avoidance gives $3! - 9 \cdot 2! + 18 \cdot 1! - 6 \cdot 0! = 0$, as every permutation uses three cells of the full board. Rook placements specify partial bad positions rather than complete permutations.

Inclusion–exclusion solutions (continued)

5 • Rectangular assignments

The three components share neither rows nor columns and contribute $1+x,1+x,1+2x$. Their product is $1+4x+5x^2+2x^3$. With $m=3,n=4$, k fixed cells leave $(4-k)!/(4-3)!$ injections. Thus $24-4 \cdot 6+5 \cdot 2-2 \cdot 1=8$ legal assignments.

In job order they are $(2,3,1),(2,4,1),(3,1,2),(3,4,1),(3,4,2),(4,1,2),(4,3,1),(4,3,2)$. No person repeats; job three goes only to 1 or 2. The denominator happens to be 1! here but must remain in the general rectangular formula.

6 • Conditions for board multiplication

Every placement C uniquely splits into $C \cap B_1$ and $C \cap B_2$. Conversely, any two component placements combine legally because neither row nor column is shared. This bijection, classified by i rooks in B_1, proves coefficient convolution and hence polynomial multiplication.

The smallest counterexample has $B_1=\{(1,1)\},B_2=\{(1,2)\}$. Each gives $1+x$; their product contains x^2. Their union permits only one rook and gives $1+2x$. The reverse combination fails, while splitting an already legal placement is still unique.

7 • Reconstruct the hit distribution

With $S_0=30$, inversion gives $E_3=2,E_2=12-3 \cdot 2=6,E_1=26-2 \cdot 12+3 \cdot 2=8,E_0=30-26+12-2=14$, totaling 30. At least two gives $H_2=6+2=8=S_2-2S_3$.

Reconstruct $S_0=14+8+6+2=30,S_1=8+2 \cdot 6+3 \cdot 2=26,S_2=6+3 \cdot 2=12,S_3=2$. Using only the first two orders gives union lower bound $S_1-S_2=14$ and upper bound $S_1=26$. Including S_3 gives exact union 16. Independence is unnecessary.

8 • Why truncated bounds have directions

Substitute $\binom{c}{k}=\binom{c-1}{k}+\binom{c-1}{k-1}$ into the alternating sum and shift the second sum; all internal terms cancel:

$$\sum_{k=0}^{r}(-1)^k\binom{c}{k}=(-1)^r\binom{c-1}{r}.$$

Out-of-range binomials are zero. The union truncation is one minus this sum, at least one for odd r and at most one for even r. For $c=0$ both truncation and union indicator are zero. Summing gives odd-order upper and even-order lower bounds; complement directions reverse. The full identity survives arbitrary real weights, but weighted inequalities require nonnegative weights.

Inclusion–exclusion solutions (continued)

For a counterexample, one object has weight −1 and satisfies two events. The union weight is −1 while $S_1=-2$. Treating first order as an upper bound asserts $-1\le -2$, falsely. The full identity $S_1-S_2=-2-(-1)=-1$ remains valid.

9 • Bounded integer solutions

Without bounds, $\binom{10}{2}=45$. A bad event $x_i\ge 4$ shifts by four, leaving total four and $\binom{6}{2}=15$ solutions each; singles sum to 45. Two violations shift by eight, leaving zero and one solution per pair, three total. Three violations require at least twelve. The answer is $45-45+3=3$.

Alternatively $y_i=3-x_i\ge 0$ gives $y_1+y_2+y_3=1$, automatically respecting upper bounds, with three choices. The x solutions are the three permutations of (2,3,3). In general $x_i>b_i$ first occurs at integer b_i+1. Shifting by b_i would wrongly classify the legal boundary as bad.

10 • Noncoprime divisors

Multiples of 4 number 15 and of 6 number 10. Both means multiples of lcm(4,6)=12, numbering five. Thus $60-15-10+5=40$. Using 24 adds back only two, giving 37 and failing to restore 12,36,60.

Distinct prime divisors of 36 are only 2,3. Coprimality is avoiding these two events, so $\phi(36)=36-18-12+6=12=36(1-1/2)(1-1/3)$. Prime powers are not separate events: divisibility by two already contains divisibility by four. The product derivation uses distinct primes.

11 • Full ladder-board distribution

$r_0=1, r_1=6$. Of fifteen cell pairs, three share a row and two share a column; distinct cells cannot share both. Hence $r_2=10$. Three rooks occupy each nonempty row, with column triples exactly (1,2,3),(1,2,4),(1,3,4),(2,3,4), so $r_3=4$. Empty row four gives $r_4=0$.

The intersection sums $r_k(4-k)!$ are [24,36,20,4,0]. Inversion gives $h_0=24-36+20-4=4, h_1=36-2\cdot 20+3\cdot 4=8, h_2=20-3\cdot 4=8, h_3=4, h_4=0$. They total 24, and $\sum_t t h_t = 8+16+12=36=S_1$.

Avoiding permutations are (3,1,2,4),(3,4,1,2),(3,4,2,1),(4,1,2,3). Coordinates correspond to the same rows, and columns occur once each. The count $h_1=8$ differs from forbidden-cell count $r_1=6$.

Inclusion–exclusion solutions (continued)

12 • Circular contraction and orientation

Contract k specified pairs to labeled blocks, leaving 8−k objects in (7−k)! circular orders. Each pair has two internal orientations, giving 2^k. At least four blocks remain, so contractions and orientations are uniquely reversible. Intersection sums are $\binom{4}{k}2^k(7-k)!$. Inclusion–exclusion gives

$$7! - 4 \cdot 2 \cdot 6! + 6 \cdot 4 \cdot 5! - 4 \cdot 8 \cdot 4! + 16 \cdot 3! = 1488.$$

Rotations are already removed by the circular factorial; reflections remain distinct. For two people the two orientations are the same rotation class, so multiplying by two repeats an arrangement. The general statement needs at least two specified pairs.

13 • No isolates versus connectivity

There are $2^{\binom{4}{2}} = 64$ graphs. Specified k isolated vertices forbid all incident edges; only edges among 4−k remaining vertices are free, giving $2^{\binom{4-k}{2}}$. No-isolate graphs number 64−4 • 8+6 • 2−4 • 1+1=41.

A disconnected graph without isolates has every component size at least two, so four vertices must split 2+2. Each two-vertex component has its unique edge. The three matchings are {12,34},{13,24},{14,23}. Connected graphs therefore number 41−3=38. Minimum degree one does not exclude these three.

14 • Rook dynamic-programming states

Let $d_i(S)$ count legal placements in rows 1,…,i using exactly S. Initially $d_0(\varnothing)=1$, others zero. Each old state either leaves this row empty or chooses a forbidden cell (i,j) with $j \notin S$ and contributes to new[$S \cup \{j\}$]. Clear new first and read all contributions only from old.

Every new placement uniquely determines whether this row is empty, otherwise its sole rook's j, and the preceding placement after removal. This proves completeness and uniqueness. After m rows, r_k sums $d_m(S)$ over |S|=k. There are O(mn2^n) integer additions and two O(2^n) layers of counts; large-count bit costs are separate.

In-place failure: start at $d(\varnothing)=1$, update $d(\{1\})=1$, then read that new value to add column two and obtain $d(\{1,2\})=1$. One row cannot contain two rooks. Separate old/new arrays preserve the previous-row state meaning.

For the three-cell board choose e=(1,1). Omitting e leaves (1,2),(2,1), two nonattacking cells with polynomial $(1+x)^2$. Selecting e removes its entire row and column, leaving empty-board polynomial one; multiply by x for e. Adding cases gives $1+3x+x^2$.

Inclusion–exclusion solutions (continued)

Verification code appendix: verify_derangements.py

```
"""Compare three derangement identities and the rook-board example."""

from itertools import permutations
from math import comb, factorial


def enumerated_derangements(n: int) -> int:
    return sum(all(i != x for i, x in enumerate(p)) for p in permutations(range(n)))


def recurrence(n: int) -> int:
    if n == 0:
        return 1
    if n == 1:
        return 0
    d0, d1 = 1, 0
    for k in range(2, n + 1):
        d0, d1 = d1, (k - 1) * (d0 + d1)
    return d1


def inclusion_exclusion(n: int) -> int:
    return sum((-1) ** k * comb(n, k) * factorial(n - k) for k in range(n + 1))


for n in range(8):
    expected = enumerated_derangements(n)
    assert recurrence(n) == expected
    assert inclusion_exclusion(n) == expected

# Three forbidden cells in a 3x3 board have r0=1, r1=3, r2=1.
avoid = 6 - 3 * 2 + 1
assert avoid == 1

print("derangement examples: PASS")
```

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study